DEPARTMENT OF PHYSICS
INDIAN INSTITUTE OF TECHNOLOGY, MADRAS

PH1020 Physics II

Problem Set 6

Magnetic vector potential and Magnetic fields in matter Date: 25 Mar 2025

1. Find the vector potential above and below the current sheet, lies in the xy-plane, with uniform current density $\boldsymbol{K} = K\hat{\mathbf{x}}$ (See example 5.8 of Griffith 4rd edition). Also, verify the magnetostatic boundary condition for the vector potential.

2. A circular loop of wire, with radius R, lies in the xy-plane, centered at the origin, and carries a current I running counterclockwise as viewed from the positive z-axis. Calculate the magnetic field of this loop assuming it to be a dipole.

3. An infinitely long cylinder of radius a has its axis along the z-axis. Its magnetization is given in cylindrical polar coordinates by $\mathbf{M} = M_0(\varrho/a)^2\hat{\boldsymbol{\varphi}}$, where M_0 is a constant. Find $\mathbf{J}_b$ and $\mathbf{K}_b$ as well as $\mathbf{B}$ and $\mathbf{H}$ both inside and outside the cylinder.

4. Consider a toroid in which a wedge-shaped region of small angle ψ is absent, as shown in the figure 1. A steady current I flows in it. The inner radius of the toroid is R, and the total number of turns in it is N. Assume that the magnetic field $\mathbf{B}$ in the air gap is still along $\hat{\boldsymbol{\varphi}}$. Find $\mathbf{H}$ in the toroid given that the core of the toroid is a LIH magnetic material with magnetic susceptibility χ_m.

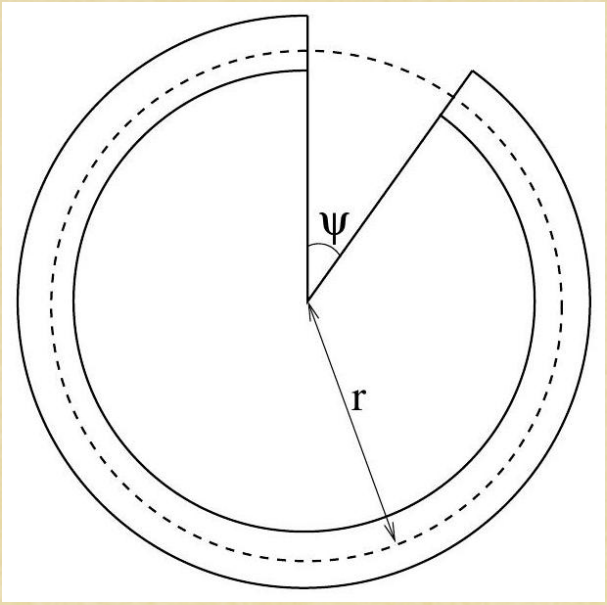


Figure 1: Top view of the toroid. (Figure for problem 4)

5. An infinite planar magnetic sheet of thickness d having a nonuniform permeability given by $\mu(z) = \mu_0[1 + (z/d)]^2$ occupies the region $0 \leq z \leq d$. There is vacuum on either side of the sheet. A magnetic field $\mathbf{B} = B_0\hat{\boldsymbol{y}}$ (where B_0 is

a constant) is applied in the entire space. The sheet has no free current on it. Find the magnetization surface current densities at $z = 0$ and $z = d$, and also the magnetization volume current density as a function of z.

6. A very long cylinder of radius R carries a magnetization $\boldsymbol{M} = ks\hat{\boldsymbol{z}}$. Here, k is a constant and s is the distance from the axis which lies along $\hat{\boldsymbol{z}}$. Find the magnetic field inside and outside the cylinder by (a) employing Ampere's law and (b) calculating the bound currents.

7. A large piece of material have "frozen-in" magnetization $\boldsymbol{M}$. The field measured inside the material is $\boldsymbol{B_0}$. Now a cavity is hollowed out of the material. Calculate the fields $\boldsymbol{B}$ and $\boldsymbol{H}$ at the centre of the cavity, if it is

   (a) a small spherical cavity,

   (b) a long needle-shaped cavity running parallel to $\boldsymbol{M}$,

   (c) a thin wafer-shaped cavity perpendicular to $\boldsymbol{M}$.

8. A coaxial cable consists of two very long cylindrical tubes, separated by linear insulating material of magnetic susceptibility χ_m. A current I flows down the inner conductor and returns along the outer one; in each case, the current distributes itself uniformly over the surface (see Figure 2). Find the magnetic field in the region between the tubes. As a check, calculate the magnetization and the bound currents, and confirm that (together, of course, with the free currents) they generate the correct field.

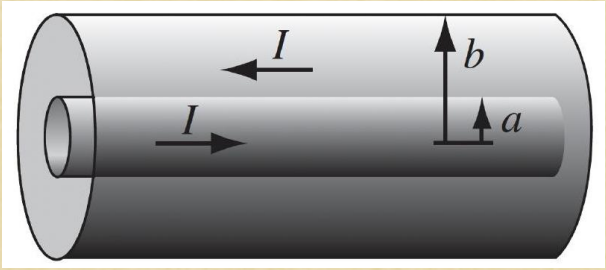


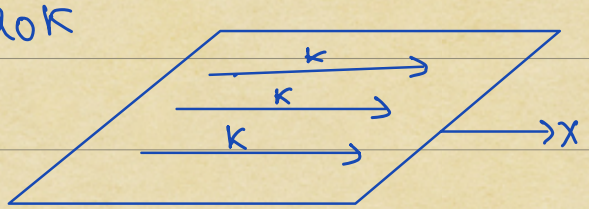
Figure 2: A coaxial cable with linear insulating material of magnetic susceptibility χ_m. Figure for problem 8.

1. Find the vector potential above and below the current sheet, lies in the xy-plane, with uniform current density $\boldsymbol{K} = K\hat{\mathbf{x}}$ (See example 5.8 of Griffith 4rd edition). Also, verify the magnetostatic boundary condition for the vector potential.

$B^{\parallel}_{\text{above}} - B^{\parallel}_{\text{below}} = \mu_0 K$

$B^{\perp}_{\text{above}} = B^{\perp}_{\text{below}}$

$\bar{A} \rightarrow$ continuous always



$(B_{top} = -\hat{y})$

$(B_{bottom} = +\hat{y})$

let $\bar{A} = A_x\hat{x} + A_y\hat{y} + A_z\hat{z} = A_x\hat{x}$

$(A_y, A_z = 0$ (along current dir^n)

$B = \frac{\mu_0 K}{2}$ (independent of distance))

$\bar{\nabla} \times \bar{A} = \bar{B}$

$$\frac{dA_x\hat{y}}{dz} = \begin{cases} -\frac{\mu_0 K}{2}\hat{y} & z>0 \\ \frac{\mu_0 K}{2}\hat{y} & z<0 \end{cases}$$

$$\Rightarrow A = \begin{cases} \left[-\frac{\mu_0 K z}{2} + f(y)\right]\hat{x} & z>0 \\ \left[\frac{\mu_0 K z}{2} + g(y)\right]\hat{x} & z<0 \end{cases}$$

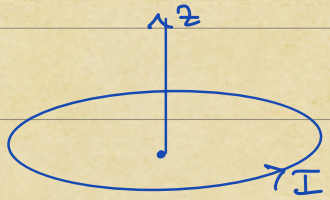
$f(y) = g(y)$ (A is continuous along z-axis)

Boundary condition: $\left.\frac{dA}{dz}\right|_{z>0} - \left.\frac{dA}{dz}\right|_{z<0} = -\mu_0 K$

2. A circular loop of wire, with radius R, lies in the xy-plane, centered at the origin, and carries a current I running counterclockwise as viewed from the positive z-axis. Calculate the magnetic field of this loop assuming it to be a dipole.

$\vec{m} = I \times \pi R^2 \hat{z}$

$\vec{B} = \frac{\mu_0}{4\pi r^3}\left[3(\vec{m}\cdot\hat{r})\hat{r} - \vec{m}\right]$



3. An infinitely long cylinder of radius a has its axis along the z-axis. Its magnetization is given in cylindrical polar coordinates by $\mathbf{M} = M_0(\varrho/a)^2\hat{\boldsymbol{\varphi}}$, where M_0 is a constant. Find $\mathbf{J}_b$ and $\mathbf{K}_b$ as well as $\mathbf{B}$ and $\mathbf{H}$ both inside and outside the cylinder.

$\vec{M} = M_0(s/a)^2\hat{\phi}$

$\vec{\nabla}\times\vec{A} = \left[\frac{1}{\rho}\frac{\partial A_z}{\partial\varphi} - \frac{\partial A_\varphi}{\partial z}\right]\hat{e}_\rho + \left[\frac{\partial A_\rho}{\partial z} - \frac{\partial A_z}{\partial\rho}\right]\hat{e}_\varphi + \frac{1}{\rho}\left[\frac{\partial}{\partial\rho}(\rho A_\varphi) - \frac{\partial A_\rho}{\partial\varphi}\right]\hat{e}_z$

$\vec{J}_b = \vec{\nabla}\times\vec{M} = \frac{1}{s}\frac{d}{ds}(M_0 s^3/a^2)\hat{z} = \frac{3M_0 s}{a^2}\hat{z}$

$\vec{K}_b = \vec{M}\times\hat{n} = -M_0\hat{z}$ (curved surface $\to \hat{r}$)

$\oint \vec{B}\cdot d\vec{l} = \mu_0 I_{enc}$ $\qquad$ $\oint \vec{H}\cdot d\vec{L} = (I_f)_{enc}$

$s<a$: $B\cdot 2\pi s = \mu_0 \int \frac{3M_0 s}{a^2} \times 2\pi s\, ds$ $\qquad$ $\Rightarrow H = 0$ everywhere

$B\cdot s = \frac{3\mu_0 M_0 s^3}{3a^2}$

$\Rightarrow B = \mu_0 M_0 \left(\frac{s}{a}\right)^2 \hat{\phi}$

$s>a$: $B = 0$ (I_{enc} must be 0)

4. Consider a toroid in which a wedge-shaped region of small angle ψ is absent, as shown in the figure 1. A steady current I flows in it. The inner radius of the toroid is R, and the total number of turns in it is N. Assume that the magnetic field $\mathbf{B}$ in the air gap is still along $\hat{\varphi}$. Find $\mathbf{H}$ in the toroid given that the core of the toroid is a LIH magnetic material with magnetic susceptibility χ_m.

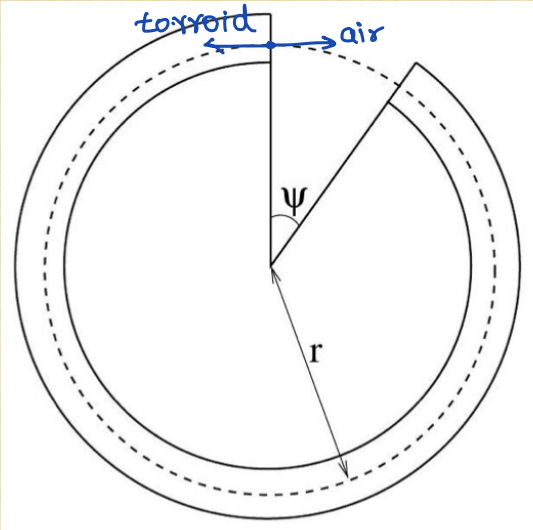


Figure 1: Top view of the toroid. (Figure for problem 4)

$\oint \bar{B} \cdot d\bar{l} = \mu_0 I_{enc}$

$\vec{B}_{toroid} = \mu_0 (1+\chi_m) \vec{H}_{toroid}$

$\vec{B}_{air\ gap} = \mu_0 \vec{H}_{air\ gap}$

Ampere's law:

$$\vec{B}_{toroid} \times r(2\pi - \Psi) + \vec{B}_{air\ gap} \times r\Psi = \mu_0 [NI]$$

$$\Rightarrow \mu_0 r[(1+\chi_m)(2\pi - \Psi)\vec{H}_{toroid} + \Psi \vec{H}_{air\ gap}] = \mu_0 NI$$

⊥ component of magnetic field is continuous

$$(1+\chi_m)\bar{H}_{toroid} = \bar{H}_{air\ gap}$$

$$\Rightarrow r(1+\chi_m) \cdot 2\pi \vec{H}_{toroid} = NI$$

$$\Rightarrow \vec{H}_{toroid} = \frac{NI}{2\pi r(1+\chi_m)}$$

5. An infinite planar magnetic sheet of thickness d having a nonuniform permeability given by $\mu(z) = \mu_0[1 + (z/d)]^2$ occupies the region $0 \le z \le d$. There is vacuum on either side of the sheet. A magnetic field $\mathbf{B} = B_0\hat{\boldsymbol{y}}$ (where B_0 is a constant) is applied in the entire space. The sheet has no free current on it. Find the magnetization surface current densities at $z = 0$ and $z = d$, and also the magnetization volume current density as a function of z.

By symmetry, H varies only along z

$$\vec{H} = H_1(z)\hat{x} + H_2(z)\hat{y} + H_3(z)\hat{z}$$

$$\vec{H} = \frac{\vec{B}_0}{\mu_0} \qquad \bar{\nabla}\times\bar{H} = 0 \Rightarrow \frac{d}{dz}H_2(z) = 0 \quad \text{(no free current)}$$

$$\frac{d}{dz}H_1(z) = 0$$

$$\vec{H} = C_1\hat{x} + C_2\hat{y} + H_3(z)\hat{z} \qquad \rightarrow H_1 = H_2 \rightarrow \text{const}$$

$$C_2 = \frac{B_0}{\mu_0} \qquad C_1 = 0$$

$$\Rightarrow \vec{H} = \frac{B_0}{\mu_0}\hat{y} + H_3(z)\hat{z}$$

$$\nabla\cdot\bar{B} = 0 \Rightarrow B_1^{inside} = B_1^{outside}$$

$$\mu(z)\,H_3(z) = \mu_0 (H)_{outside} = 0$$

$$\Rightarrow H_3(z) = 0$$

$$\Rightarrow \bar{H} = \frac{B_0}{\mu_0}\hat{y}$$

$$B = B_0\hat{y} \quad \text{(outside)}$$

$$= B_0(1+z/d)^2\hat{y} \quad \text{(inside)}$$

$$M = \frac{B}{\mu_0} - H = 0 \quad \text{outside}$$

$$= \frac{B_0}{\mu_0}\left[\frac{z^2}{d^2} + \frac{2z}{d}\right]\hat{y}$$

$$J_b = \bar{\nabla} \times \bar{m} = -\hat{x}\left(\frac{dm_y}{dz}\right)$$

$$= -\left(\frac{2z}{d^2} + \frac{2}{d}\right)\frac{B_0}{\mu_0}\hat{y}$$

$$K_b = \bar{M} \times \hat{n} = \frac{3B_0}{\mu_0}\hat{x} \quad \text{(top)}$$

$$= 0 \quad \text{(bottom)}$$

6. A very long cylinder of radius R carries a magnetization $\boldsymbol{M} = ks\hat{\boldsymbol{z}}$. Here, k is a constant and s is the distance from the axis which lies along $\hat{\boldsymbol{z}}$. Find the magnetic field inside and outside the cylinder by (a) employing Ampere's law and (b) calculating the bound currents.

(a) $H_{in}(L) - H_{out}(L) = 0$

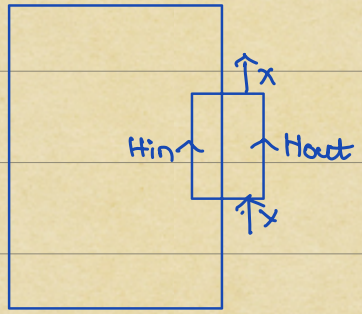
$H_{out} = H_{in} = 0$

$M_{out} = 0$

$B_{out} = 0$

$$H_{in} = \frac{B_{in}}{\mu_0} - M_{in}$$

$$\Rightarrow B_{in} = \mu_0 ks\hat{z}$$



(b) $K_b = \bar{m} \times \hat{n}$ $\quad J_b = \bar{\nabla} \times \bar{m}$

$= (ks\hat{z}) \times \hat{r}$ $\quad = -k\hat{\phi}$

$= kR\hat{\phi}$

due to K_b → Solenoid $\quad \bar{B}_{out} = 0$

$$\bar{B}_{inside} = \mu_0 K_b = \mu_0 KR$$

due to circulating inside current:

$$B \cdot 2\pi s = \mu_0 \int \bar{J}_b \cdot d\bar{a} + \mu_0 \int \bar{K}_b \cdot d\bar{l}$$

$$B l = \mu_0 \int -K\hat{\phi} \cdot (ds\, dz\, \hat{\phi}) + \mu_0 \int (KR\hat{\phi})\, dz$$

$$(\text{outside}) = \mu_0 (-K) R(l) + \mu_0 (KR)(L)$$

$$B_{out} = 0$$

$$B_{in} = -\mu_0 KS$$

7. A large piece of material have "frozen-in" magnetization $\boldsymbol{M}$. The field measured inside the material is $\boldsymbol{B_0}$. Now a cavity is hollowed out of the material. Calculate the fields $\boldsymbol{B}$ and $\boldsymbol{H}$ at the centre of the cavity, if it is
   (a) a small spherical cavity,
   (b) a long needle-shaped cavity running parallel to $\boldsymbol{M}$,
   (c) a thin wafer-shaped cavity perpendicular to $\boldsymbol{M}$.

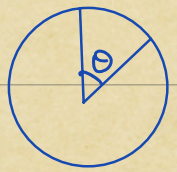
let $\bar{M} = M\hat{z}$

(a) $B = \frac{2}{3}\mu_0 M$

$$B_{net} = B_0 - \frac{2}{3}\mu_0 M$$

$$H = \frac{B}{\mu_0} = \frac{B_0}{\mu_0} - \frac{2}{3}M$$

$$= H_0 + \frac{M}{3}$$





(b) $B = \mu_0 K$ (solenoid) $= \mu_0 M$

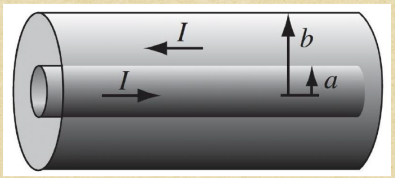
$$B_{net} = B_0 - \mu_0 M$$

$$H = \frac{B}{\mu_0} = \frac{B_0}{\mu_0} - M = H_0$$

(c) Bound currents are small and far away from center $\Rightarrow B = B_0$

$$H = \frac{B}{\mu_0} = H_0 + M$$

8. A coaxial cable consists of two very long cylindrical tubes, separated by linear insulating material of magnetic susceptibility χ_m. A current I flows down the inner conductor and returns along the outer one; in each case, the current distributes itself uniformly over the surface (see Figure 2). Find the magnetic field in the region between the tubes. As a check, calculate the magnetization and the bound currents, and confirm that (together, of course, with the free currents) they generate the correct field.



$$M = \chi_m H$$

$$= \frac{\chi_m I}{2\pi s}\hat{\phi}$$

$s \phi z$

$$H \times 2\pi s = I \Rightarrow \bar{H} = \frac{I}{2\pi s} \Rightarrow B = \mu_0(1+\chi_m)\frac{I}{2\pi s}\hat{\phi}$$

$$K_b = \bar{M} \times \hat{n} = \frac{\chi_m I}{2\pi a}\left[\hat{\phi} \times (-\hat{s})\right] = \frac{\chi_m I}{2\pi a}\hat{z}$$

$$= \frac{\chi_m I}{2\pi b}\left[\hat{\phi} \times \hat{s}\right] = -\frac{\chi_m I}{2\pi b}\hat{z}$$

$J_b = \nabla \times \bar{m} = 0.$

$$B \cdot 2\pi s = \mu_0\left[I + \frac{\chi_m I}{2\pi a} \cdot 2\pi a\right]$$

$$\Rightarrow B = \frac{\mu_0 I(1+\chi_m)}{2\pi s}\hat{\phi}$$

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Problem Set 6 - Solutions

Magnetic vector potential and Magnetic fields in matter Date: 25 Mar 2025

1. Problem-1

Question: Find the vector potential above and below the current sheet, lies in the xy-plane, with uniform current density $\boldsymbol{K} = K\hat{\mathbf{x}}$ (See example 5.8 of Griffith 4rd edition). Also, verify the magnetostatic boundary condition for the vector potential.
Solution:

Let the magnetic vector potential

$$\mathbf{A} = A_x\hat{\mathbf{x}} + A_y\hat{\mathbf{y}} + A_z\hat{\mathbf{z}}$$

From equation (5.66) of Griffiths' book it is evident that for the given system we have $A_y = A_z = 0$ ($\mathbf{A}$ is parallel(antiparallel) to $\mathbf{K}$).

Hence, A_x is the only non zero component. Now,

$$\nabla \times \mathbf{A} = \mathbf{B}$$

$$\frac{\partial A_x}{\partial z}\hat{\mathbf{y}} = \begin{cases} -\dfrac{\mu_0 K}{2}\hat{\mathbf{y}} & z > 0 \\ +\dfrac{\mu_0 K}{2}\hat{\mathbf{y}} & z < 0 \end{cases}$$

$$\mathbf{A} = \begin{cases} \left(-\dfrac{\mu_0 K}{2}z + f(y)\right)\hat{\mathbf{x}} & z > 0 \\ \left(\dfrac{\mu_0 K}{2}z + f(y)\right)\hat{\mathbf{x}} & z < 0 \end{cases}$$

Verifying the magnetostatistic boundary condition

$$\left.\frac{\partial \mathbf{A}}{\partial z}\right|_{z>0} - \left.\frac{\partial \mathbf{A}}{\partial z}\right|_{z<0} = -\mu_0\mathbf{K}$$

2. Problem-2

Question: A circular loop of wire, with radius R, lies in the xy-plane, centered at the origin, and carries a current I running counterclockwise as viewed from the positive z-axis. Calculate the magnetic field of this loop assuming it to be a dipole.
Solution:

The dipole moment of the system is

$$\mathbf{m} = I\pi R^2 \hat{\mathbf{z}}$$

Given the dipole moment of a system the magnetic field is given by (Equation (5.89) of Griffith's book)

$$\mathbf{B} = \frac{\mu_0}{4\pi}\frac{1}{r^3}[3(\mathbf{m}\cdot\hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{m}]$$

Substituting $\hat{\mathbf{z}} = \cos\theta\hat{\mathbf{r}} - \sin\theta\hat{\theta}$

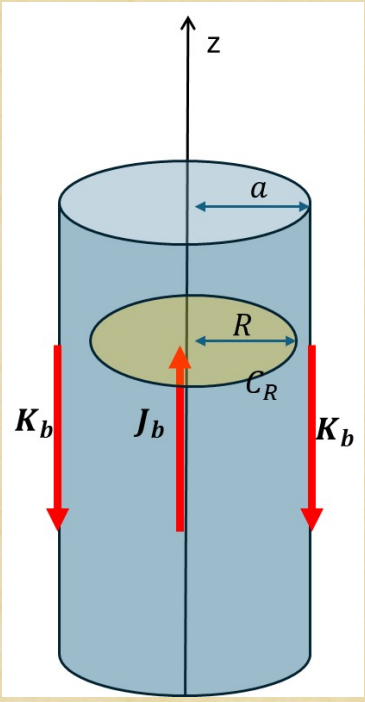
$$\mathbf{B} = \frac{\mu_0}{4\pi}\frac{I\pi R^2}{r^3}[2\cos\theta\hat{\mathbf{r}} + \sin\theta\hat{\theta}]$$

3. Problem-3

Question: An infinitely long cylinder of radius a has its axis along the z-axis. Its magnetization is given in cylindrical polar coordinates by $\mathbf{M} = M_0(\varrho/a)^2\hat{\boldsymbol{\varphi}}$, where M_0 is a constant. Find $\mathbf{J}_b$ and $\mathbf{K}_b$ as well as $\mathbf{B}$ and $\mathbf{H}$ both inside and outside the cylinder.

Solution:

We are given $\mathbf{M} = M_0\frac{Q^2}{a^2}\hat{\varphi}$ inside the cylinder.



Inside the cylinder, we have

$$\mathbf{J}_b = \boldsymbol{\nabla}\times\mathbf{M} = \frac{3M_0\varrho}{a^2}\hat{\mathbf{z}}.$$

Outside the cylinder, $\mathbf{J}_b = 0$. The bound surface current is given by

$$\mathbf{K}_b = \mathbf{M}\times\hat{\varrho}\big|_{\varrho=a} = -M_0\hat{\mathbf{z}}.$$

Since $\mathbf{J}_f = 0$ in this problem, we can determine $\mathbf{B}$ using the bound currents that we just determined. Symmetry considerations imply that we can choose $\mathbf{B} = B(\varrho)\hat{\varphi}$. To determine, $B(\varrho)$, we choose as an Amperian loop a circle of radius R centered about the z-axis and lying in a plane given by $z =$ constant. Using Ampère's law, for $R < a$, we get

$$\int_{C_R} \mathbf{B} \cdot \, \mathrm{dl} = (2\pi R)B(R) = \mu_0 \int_0^R (\mathbf{J}_b(\varrho) \cdot \hat{\mathbf{z}})\, 2\pi\varrho d\varrho,$$
$$= \frac{2\pi\mu_0 M_0 R^3}{a^2}.$$

This implies that $B(\varrho) = \frac{\mu_0 M_0 \varrho^2}{a^2}$ for $\varrho < a$. For $R > a$, a similar computation gives

$$\int_{C_R} \mathbf{B} \cdot \mathbf{dl} = (2\pi R)B(R) = \mu_0 \int_0^a \mathbf{J}_b(\varrho) \cdot \hat{\mathbf{z}} 2\pi\varrho d\varrho + \mu_0(2\pi a)\,(-M_0)\,,$$
$$= \frac{2\pi\mu_0 M_0 a^3}{a^2} + \mu_0(2\pi a)\,(-M_0)$$
$$= 0 \Longrightarrow B(R) = 0.$$

We thus obtain

$$\mathbf{B} = \begin{cases} \frac{\mu_0 M_0 \rho^2}{a^2}\hat{\varphi} & \text{for } \varrho < a, \\ 0 & \text{for } \varrho > a. \end{cases}$$

The discontinuity in $\mathbf{B}$ at $\varrho = a$ is expected due to the presence of a non-zero surface current. The student is asked to check that the discontinuity is as expected i.e.

$$\hat{n} \times (\mathbf{B}_2 - \mathbf{B}_1) = \mu_0 \mathbf{K}_b.$$

We determine $\mathbf{H}$ using the relation $\mathbf{H} = \frac{\mathbf{B}}{\mu_0} - \mathbf{M}$ to obtain

$$\mathbf{H} = 0 \text{ everywhere.}$$

The vanishing of $\mathbf{H}$ is related to the absence of free currents.

4. Problem-4

Question: Consider a toroid in which a wedge-shaped region of small angle ψ is absent, as shown in the figure 1. A steady current I flows in it. The inner radius of the toroid is R, and the total number of turns in it is N. Assume that the magnetic field $\mathbf{B}$ in the air gap is still along $\hat{\boldsymbol{\varphi}}$. Find $\mathbf{H}$ in the toroid given that the core of the toroid is a LIH magnetic material with magnetic susceptibility χ_m.
Solution:

Recall that for a toroidal coil, there is a constant auxiliary field $\mathbf{H}$ (and $\mathbf{B}$) directed along $\hat{\varphi}$ in its interior. The magnitude is determined in terms of the current and the total number of coils. We assume that the magnetic field $\mathbf{H}$ (and $\mathbf{B}$) is along $\hat{e}_\varphi$ - this is a good approximation if the wedge angle ψ is small. At the boundary of the wedge, the normal component of $\mathbf{B}$ must be continuous. Since the normal to the wedge is along $\hat{\varphi}$, we have $\mathbf{B}_{\text{air}} = \mathbf{B}_{\text{toroid}}$. Thus, the auxiliary field is given by

$$\mathbf{B}_{\text{toroid}} = \mu_0 (1 + \chi_m)\, \mathbf{H}_{\text{toroid}} \quad , \quad \mathbf{B}_{\text{air}} = \mu_0 \mathbf{H}_{\text{air}}$$

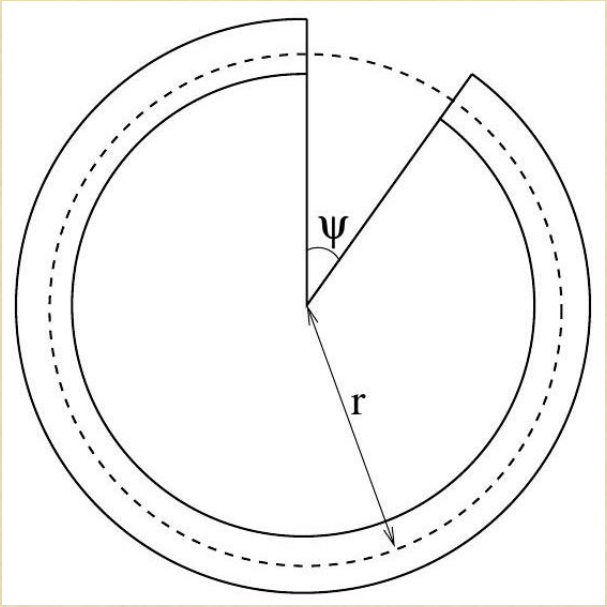


Figure 1: Top view of the toroid.

The continuity of the normal component of **B** implies that

$$\mathbf{H}_{\text{air}} = (1+\chi_m)\,\mathbf{H}_{\text{toroid}}\,.$$

In order to determine **H**, we choose an Amperian loop of radius r (indicated by a dashed line in Figure 1) passing through the interior of the toroid. Carrying out the line integral over **H**, we get

$$\begin{aligned} NI &= H_{\text{toroid}}\,(2\pi-\psi)r + H_{\text{air}}\psi r \\ &= H_{\text{toroid}}\,r\,(2\pi-\psi+(1+\chi_m)\,\psi) \\ &= H_{\text{toroid}}\,r\,(2\pi+\chi_m\psi) \\ \implies \mathbf{H}_{\text{toroid}} &= \frac{NI}{(2\pi+\chi_m\psi)\,r}\hat{\boldsymbol{\varphi}} \end{aligned}$$

5. Problem-5

Question: An infinite planar magnetic sheet of thickness d having a nonuniform permeability given by $\mu(z)=\mu_0[1+(z/d)]^2$ occupies the region $0\le z\le d$. There is vacuum on either side of the sheet. A magnetic field $\mathbf{B}=B_0\hat{\boldsymbol{y}}$ (where B_0 is a constant) is applied in the entire space. The sheet has no free current on it. Find the magnetization surface current densities at $z=0$ and $z=d$, and also the magnetization volume current density as a function of z.

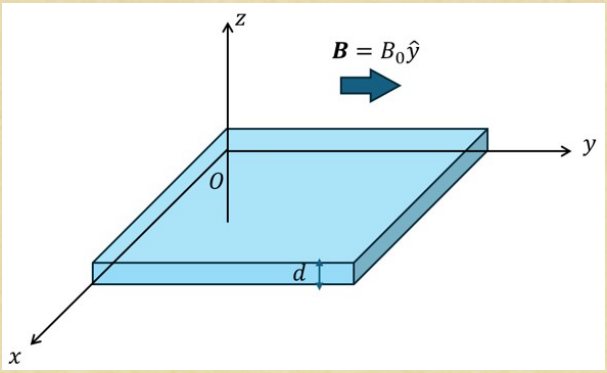
Solution:

Translation invariance in the x and y directions implies that

$$\mathbf{H}=\mathbf{H}(z)=H_1(z)\hat{\mathbf{x}}+H_2(z)\hat{\mathbf{y}}+H_3(z)\hat{\mathbf{z}}.$$

The boundary conditions on $H(z)$ is

$$\lim_{|z|\to\infty} H(z)=\frac{B_0}{\mu_0}.$$



But $(\boldsymbol{\nabla} \times \mathbf{H}) = 0$ implies that $dH_i(z)/dz = 0$ or $H_i(z) =$ constant for $i = 1, 2$ with no condition on $H_3(z)$. We thus obtain

$$\mathbf{H} = \frac{B_0}{\mu_0}\hat{\mathbf{y}} + H_3(z)\hat{\mathbf{z}} \quad \text{everywhere .}$$

To fix $H_3(z)$, we need to impose $\boldsymbol{\nabla} \cdot \mathbf{B} = 0$. Outside the strip it implies $H_3(z)$ is a a constant which vanishes due to the boundary conditions as $|z| \to \infty$. This implies that

$$\mathbf{H} = \frac{B_0}{\mu_0}\hat{\mathbf{y}} \quad \text{outside the strip.}$$

Inside the strip, $\boldsymbol{\nabla} \cdot \mathbf{B} = 0$ implies that $\mu(z)H_3(z)$ is a constant. Continuity of the z component of $\mathbf{B}$, at $z = 0$, forces this constant to vanish. Thus, we obtain that $H_3(z) = 0$ everywhere.

$$\mathbf{H} = \frac{B_0}{\mu_0}\hat{\mathbf{y}} \quad \text{everywhere .}$$

From this we obtain the magnetic field to be

$$\mathbf{B} = \begin{cases} B_0\left(1 + \frac{z}{d}\right)^2 \hat{\mathbf{y}}, & \text{for } 0 < z < d, \\ B_0\hat{\mathbf{y}}, & \text{outside the magnetic sheet .} \end{cases}$$

The magnetization is determined using $\mathbf{M} = \frac{\mathbf{B}}{\mu_0} - \mathbf{H}$ and we get

$$\mathbf{M} = \begin{cases} \frac{B_0}{\mu_0}\left(2\frac{z}{d} + \frac{z^2}{d^2}\right)\hat{\mathbf{y}}, & \text{for } 0 < z < d \\ 0, & \text{outside the magnetic sheet} \end{cases}$$

We can determine the current densities using the formulae $\mathbf{J}_b = \boldsymbol{\nabla} \times \mathbf{M}$ and $\mathbf{K}_b = (\mathbf{M} \times \hat{n})$. We get that volume charge density for $0 < z < d$ is given by

$$\mathbf{J}_b = -\frac{2B_0}{\mu_0 d}\left(1 + \frac{z}{d}\right)\hat{\mathbf{x}}$$

Using $(\mathbf{M} = 0, \hat{n} = -\hat{\mathbf{z}})$ at $z = 0$ and $(\mathbf{M} = (3B_0/\mu_0)\,\hat{\mathbf{y}}, \hat{n} = \hat{\mathbf{z}})$ at $z = d$, the surface current densities are

$$\mathbf{K}_b = \begin{cases} 0, & \text{at } z = 0, \\ \frac{3B_0}{\mu_0}\hat{\mathbf{x}}, & \text{at } z = d. \end{cases}$$

Again, the student is asked to check the consistency of the expressions for $\mathbf{B}$ and $\mathbf{K}_b$ by checking that the discontinuity at the two interfaces are consistent with $\hat{n} \times (\mathbf{B}_2 - \mathbf{B}_1) = \mu_0\mathbf{K}_b$.

6. Problem-6

Question: A very long cylinder of radius R carries a magnetization $\boldsymbol{M} = ks\hat{\boldsymbol{z}}$. Here, k is a constant and s is the distance from the axis which lies along $\hat{\boldsymbol{z}}$. Find the magnetic field inside and outside the cylinder by (a) employing Ampere's law and (b) calculating the bound currents.

Solution:

(a) It is obvious from the symmetry that $\boldsymbol{H}$ directs along $\hat{\boldsymbol{z}}$.
Using the Amperian loop as in Figure 2

$$\oint \boldsymbol{H} \cdot d\boldsymbol{l} = I_{\text{fenc}}$$

$Hl = 0$ (the right hand side is zero as there is no free current)

Therefore, $\boldsymbol{H} = 0$. Using equation (6.18) of Griffith's book $\left(\boldsymbol{H} \equiv \frac{1}{\mu_0}\boldsymbol{B} - \boldsymbol{M}\right)$ we get,

$$\boldsymbol{B}_{\text{outside}} = 0$$
$$\boldsymbol{B}_{\text{inside}} = \mu_0 ks\hat{\boldsymbol{z}}$$

(b) Given the magnetization vector $\boldsymbol{M} = ks\hat{\boldsymbol{z}}$, the volume bound current is

$$J_b = \nabla \times M = -k\hat{\phi}$$

and the surface bound current is

$$\boldsymbol{K}_b = \boldsymbol{M} \times \hat{\boldsymbol{n}} = kR\hat{\boldsymbol{\phi}}$$

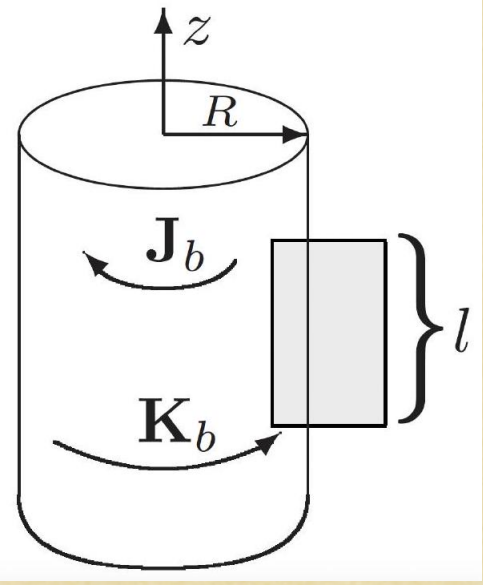


Figure 2: The cylinder with volume bound current $\boldsymbol{J}_b$ and surface bound current $\boldsymbol{K}_b$. The Amperian loop for of length l is shown.

It is evident from the bound current computation that the system is basically superposition of two solenoids. Therefore, [1]

[03] Students are encouraged to revisit Example 5.9 of Griffith's book.

$$\boldsymbol{B}_{\text{outside}} = 0$$

Now for inside the cylinder,

$$\oint \boldsymbol{B} \cdot d\boldsymbol{l} = \mu_0 \left[\int J_b da + \int K_b dl \right]$$
$$Bl = \mu_0[-k(R-s)l + kRl]$$

Therefore,

$$\boldsymbol{B}_{\text{inside}} = \mu_0 ks\hat{\boldsymbol{z}}.$$

7. Problem-7

Question: A large piece of material have "frozen-in" magnetization $\boldsymbol{M}$. The field measured inside the material is $\boldsymbol{B_0}$. Now a cavity is hollowed out of the material. Calculate the fields $\boldsymbol{B}$ and $\boldsymbol{H}$ at the centre of the cavity, if it is

(a) a small spherical cavity,

(b) a long needle-shaped cavity running parallel to $\boldsymbol{M}$,

(c) a thin wafer-shaped cavity perpendicular to $\boldsymbol{M}$.

Solution:

Hollowing out of a material is equivalent to filling the cavity with same material and material with negative magnetization. Remember, we have followed the similar kind of approach while solving such "cavity problems" in electrostatics.

(a) The magnetic field of a uniformly magnetized sphere is $\frac{2}{3}\mu_0\boldsymbol{M}$ (refer equation 6.16 of Griffith's book). If we remove the sphere, the magnetic field is

$$\boldsymbol{B} = \boldsymbol{B_0} - \frac{2}{3}\mu_0\boldsymbol{M}$$

Similarly,

$$\begin{aligned}\boldsymbol{H} &= \frac{1}{\mu_0}\boldsymbol{B} \\ &= \frac{1}{\mu_0}\left[\boldsymbol{B_0} - \frac{2}{3}\mu_0\boldsymbol{M}\right] \\ &= \boldsymbol{H_0} + \frac{1}{3}\boldsymbol{M}\end{aligned}$$

(b) The needle can be considered as a long solenoid. The field inside the long solenoid is $\mu_0\boldsymbol{K}_b$. But we have just seen in the previous problem that $|\boldsymbol{K}_b| = |\boldsymbol{M}|$, so magnetic field due to the bound current on the inside surface of the cavity is $\mu_0\boldsymbol{M}$, pointing downward. Therefore,

$$\boldsymbol{B} = \boldsymbol{B}_0 - \mu_0\boldsymbol{M}$$

Then,

$$\begin{aligned}\boldsymbol{H} &= \frac{1}{\mu_0}\boldsymbol{B} \\ &= \mu_0\left(\boldsymbol{B_0} - \mu_0\boldsymbol{M}\right) \\ &= \boldsymbol{H}_0\end{aligned}$$

(c) In this case, the bound currents are small and far away from the center so

$$\boldsymbol{B} = \boldsymbol{B}_0$$

It is easy to check $\boldsymbol{H} = \boldsymbol{H_0} + \boldsymbol{M}$ Comment: In the wafer, $\boldsymbol{B}$ is the field in the medium. In the needle, $\boldsymbol{H}$ is $\boldsymbol{H}$ in the medium. Both $\boldsymbol{B}$ and $\boldsymbol{H}$ are modified in the sphere.

8. Problem-8

Question: A coaxial cable consists of two very long cylindrical tubes, separated by linear insulating material of magnetic susceptibility χ_m. A current I flows down the inner conductor and returns along the outer one; in each case, the current distributes itself uniformly over the surface (see Figure 3). Find the magnetic field in the region between the tubes. As a check, calculate the magnetization and the bound currents, and confirm that (together, of course, with the free currents) they generate the correct field.

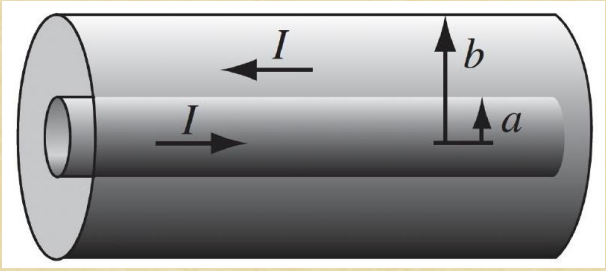


Figure 3: A coaxial cable with linear insulating material of magnetic susceptibility χ_m.

Solution:
Using Ampere's law,

$$\oint \boldsymbol{H}\cdot d\boldsymbol{l} = I_{f_{enc}}$$

$$H2\pi s = I$$

Therefore,

$$\boldsymbol{H} = \frac{I}{2\pi s}\hat{\boldsymbol{\phi}}, \quad a < s < b$$

Now, using equation (6.30) of Griffith's book,

$$\boldsymbol{B} = \mu_0\left(1+\chi_m\right)\boldsymbol{H} = \mu_0\left(1+\chi_m\right)\frac{I}{2\pi s}\hat{\boldsymbol{\phi}}.$$

Let's compute magnetic field by obtaining the bound currents. Given the field $\boldsymbol{H}$ the magnetization can be determined using equation (6.29) of the Griffith's book,

$$\boldsymbol{M} = \chi_m\boldsymbol{H} = \frac{\chi_m I}{2\pi s}\hat{\boldsymbol{\phi}}$$

Now,

$$\boldsymbol{J_b} = \boldsymbol{\nabla} \times \boldsymbol{M} = 0$$

$$\boldsymbol{K_b} = \boldsymbol{M} \times \hat{\boldsymbol{n}} = \begin{cases} \dfrac{\chi_m I}{2\pi a}\hat{\boldsymbol{z}}, & \text{if } s = a \\ \dfrac{\chi_m I}{2\pi b}\hat{\boldsymbol{z}}, & \text{if } s = b \end{cases}$$

The total enclosed current inside the Amperian loop considered between the cylinders,

$$I_{\text{enc}} = I + \frac{\chi_m I}{2\pi a} \cdot 2\pi a = (1 + \chi_m)\, I$$

Thus, using the Ampere's law

$$\boldsymbol{B} = \mu_0 \left(1 + \chi_m\right) \frac{I}{2\pi s}\hat{\boldsymbol{\phi}}.$$

[1]

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